

Payroll System Requirements Document

System Architecture, Legacy Spreadsheet Breakdown, and Development Guidelines

Overview

This document outlines the technical requirements, architectural mapping, and implementation guidelines for developing an automated payroll system based on an existing manual spreadsheet architecture. The legacy process relies heavily on multi-tab manual data entry, exposing operations to human error, broken references, and high administrative overhead.

The objective of the new system is to transition from static, sheet-based calculations to a unified database-driven application with automated payroll period generation, rule-based calculation engines, dynamic payslip rendering, and granular permission controls.

1. Legacy Spreadsheet Analysis & Mapping

The legacy workbook operates as an unnormalized, flat-file pseudo-database across eight interconnected sheets. The operational flow and target system replacements are defined below.

Sheet Name	Core Functionality	Primary Data Points	Legacy Limitations	Target Module
ID	Employee Master Data & Static Rates	NO . KARYAWAN, NAMA KARYAWAN, JABATAN, GAJI POKOK, OPERASIONAL, INSENTIF, T.J. JABATAN, Session Rates	Manual rate updates; no audit history for pay adjustments.	Employee Management

Sheet Name	Core Functionality	Primary Data Points	Legacy Limitations	Target Module
TEMP	Payroll Calendar Template	Cutoff window (27th to 26th), Date-to-Day dynamic mapping	Monthly manual tab duplicating and formula adjustments.	Payroll Period Engine
ABS	Attendance Logging & Deductions	HADIR, TLBT, IZIN, SAKIT, CUTI, ALFA, Fine calculations	Manual entry prone to typos; static deduction rules.	Attendance Module
AUTO INS	Variable Task Entry	Session counts (FT KLINIK, SM KLINIK, FT VISIT, etc.)	Isolated input grid detached from real-time validations.	Variable Activity Log
INS	Incentive Computation	Formula multiplication: Quantity (AUTO INS) × Rate (ID)	Broken links if rows are reordered; hidden hardcoded formulas.	Calculation Engine
DENDA	Ad-hoc Deductions & Fines	NOMINAL, KETERANGAN, Employee ID	Risk of unmapped manual entries; missing approval logs.	Adjustments Ledger
GAJI	Master Salary Reconciliation	Net total consolidation (GAJI POKOK + Allowances + Incentives - Deductions)	Slow processing; no inline validation checks before payout.	Payroll Workbench
SLIP	Printable Payslip Generation	Template populated via cell lookup formulas	Fixed layout displays zero-value lines; manual print execution.	Document Generator

Detailed Breakdown of Legacy Sheets

1.1 Sheet: ID (Employee Master Data)

- **Purpose:** Serves as the primary source of truth for employee identity and contractual pay structures.

- **Key Fields:** NO. KARYAWAN (Primary Key), NAMA KARYAWAN, ID (Alias), JABATAN (Role), GAJI POKOK (Base Salary), OPERASIONAL (Operational Allowance), INSENTIF (Base Incentive), TJ. JABATAN (Role Allowance), and activity rates (PT VISIT, ST VISIT, TA VISIT).
- **System Requirement:** Database table with historical tracking. Modifications to base pay or rates must preserve historical values so past payroll runs remain reproducible.

1.2 Sheet: TEMP (Payroll Template & Attendance Master)

- **Purpose:** Establishes the operational date range for the monthly cycle (27th of Month *N-1* through 26th of Month *N*).
- **Key Fields:** Workday totals, calendar dates, localized day headers (KAM, JUM, SAB).
- **System Requirement:** Automated period builder. Selecting a target month/year must generate the exact date sequence, tag weekends and national holidays, and map active employees automatically.

1.3 Sheet: ABS (Attendance Tracking)

- **Purpose:** Records daily attendance status and calculates preliminary attendance penalties.
- **Key Fields:** Daily status codes, aggregate totals (HADIR, TLBT, IZIN, SAKIT, CUTI, ALFA), late arrival deductions.
 - **H = present**
 - **Number = minutes late**
 - **I = not present with notice**
 - **S = not present because of sick**
 - **C = paid leave**
 - **A = not present without notice**
- **System Requirement:** Attendance entry dashboard with batch options, threshold validation, and rule-based penalty execution (e.g., automatic deduction calculation per late instance or unexcused absence).

1.4 Sheet: AUTO INS (Auto Incentive Input)

- **Purpose:** Input sheet for billable clinical activities and external site visits performed by staff.

- **Key Fields:** Activity counts categorized by venue type (INSENTIF FT KLINIK, INSENTIF SM KLINIK, INSENTIF FT VISIT).
- **System Requirement:** Variable entry form organized by clinical operational units, equipped with input bounds checking to prevent accidental high-volume data entry errors. Automated data fetch from session tracking system into the payroll system.

1.5 Sheet: INS (Incentive Calculation)

- **Purpose:** Backend matrix processing calculated earnings per task.
- **Key Fields:** Aggregate incentive values per employee per category.
- **System Requirement:** Abstracted backend processing layer executing calculations through transactional DB queries rather than volatile sheet formulas.

1.6 Sheet: DENDA (Fines/Deductions)

- **Purpose:** Manual logging of discretionary or disciplinary financial adjustments.
- **Key Fields:** NOMINAL (Amount), KETERANGAN (Reason/Description), Employee ID reference.
- **System Requirement:** Structured adjustments ledger supporting both credits (bonuses) and debits (fines), requiring explicit categories and internal audit notes.

1.7 Sheet: GAJI (Total Salary Summary)

- **Purpose:** Consolidation report showing final gross-to-net salary figures.
- **Key Fields:** Base Pay + Fixed Allowances + Variable Incentives - Attendance Deductions - Ad-hoc Fines = TOTAL GAJI (Net Salary).
- **System Requirement:** Interactive reconciliation grid providing drill-down visibility into each sub-total prior to lock and approval.

1.8 Sheet: SLIP (Payslip Generation)

- **Purpose:** Customer-facing salary documentation for staff distribution.
 - **Key Fields:** Employee details, earnings summary, itemized deductions, net payout, approval markers.
 - **System Requirement:** Dynamic layout renderer that omits zero-value line items, builds standardized PDFs, and supports queued email distribution.
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2. Developer Action Plan & Architecture Guidelines

2.1 Technical Core Modules

Module 1: Employee Data Management (CRUD & Rate Tables)

- Implement relational data models separating identity details from financial structures.
- Maintain compensation history tables to track changes in GAJI POKOK or specific visit rates without altering historical payroll runs.
- Enforce uniqueness on NO. KARYAWAN (Employee Identifier).

Module 2: Automated Period Manager

- Construct an engine that accepts configuration inputs (e.g., Start Day = 27, End Day = 26).
- Automatically instantiate monthly time buckets, resolving leap years and varying month lengths without user intervention.
- Generate baseline attendance grids pre-populated with active, non-terminated employees upon period instantiation.

Module 3: Variable Activity & Attendance Capture

- Build consolidated web forms replacing the separate entry flows for ABS, AUTO INS, and DENDA.
- Support aggregate or daily entry modes for attendance and session counts.
- Apply client-side and server-side validation to ensure numerical entries fall within reasonable operating parameters.

Module 4: Calculation Engine (Execution Order)

- The system must calculate net pay using a strict evaluation sequence:

$$Gross\ Earnings = Base\ Salary + Fixed\ Allowances + \sum (Task\ Quantity \times Task\ Rate)$$

$$Total\ Deductions = Attendance\ Penalties + Ad-hoc\ Fines$$

$$Net\ Pay = Gross\ Earnings - Total\ Deductions$$

- Decouple calculation logic from the UI using backend calculation services. Formula variables must read from system configuration tables to allow adjustments without modifying source code.

Module 5: Dynamic PDF & Document Renderer

- Implement a template rendering engine for payslips.
 - **Conditional Rendering:** Line items where the calculated value equals zero must be stripped from the visual document to ensure clear presentation.
 - Batch render finalized records into encrypted PDFs for output and distribution.
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3. Workflow Specifications

[1. PERIOD INITIALIZATION]

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[2. VARIABLE DATA CAPTURE] —▶ (Attendance, Task Counts, Adjustments)

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[3. AUTOMATED CALCULATION] —▶ (Apply Rate Tables & Deduction Rules)

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[4. RECONCILIATION & REVIEW] —▶ (HR Review & Final Lock)

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[5. PAYSLIP GENERATION] —▶ (PDF Batch Render & Distribution)

3.1 Detailed Process Description

1. **Period Initialization:** HR opens a new period by selecting the month and year. The system builds the calendar grid, flags official non-working days, and binds active staff profiles.
 2. **Variable Data Capture:** HR inputs shift counts, late flags, clinic/visit session totals, and any manual fines or bonuses. System validates inputs against expected ranges.
 3. **Automated Calculation:** The engine computes variable components against configured rate rules, generating a provisional payroll ledger.
 4. **Reconciliation & Review:** HR inspects the primary reconciliation view (GAJI equivalent). Discrepancies are fixed by modifying source inputs, which triggers an automated recalculation.
 5. **Finalization & Distribution:** HR locks the period to prevent further edits. The system creates immutable payroll records, generates individual PDF payslips with non-zero line items, and queues email notifications.
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4. Implementation Guidelines & Key Considerations

4.1 System Configuration Options

- **Flexible Rate Structure:** Store session values (PT VISIT, ST VISIT, etc.) and late deduction penalties inside configurable rate tables accessible through an administrative control panel. Hardcoding financial variables in application source code is strictly prohibited.
- **Custom Deduction Rules:** Support both percentage-based and fixed-value deduction formulas across custom penalty types.

4.2 Data Integrity & Security

- **Historical Reproducibility:** Updating an employee's current pay rate must not alter past payroll runs. Use snapshot logging or timestamped rate records for historical accuracy.
- **Role-Based Access Control (RBAC):** Restrict system access based on user roles (e.g., HR Data Clerk can enter attendance, Payroll Specialist can edit compensation details, Finance Director approves finalized payrolls).
- **Audit Logging:** Log every adjustment, override, or edit during payroll processing, recording the user ID, timestamp, target field, and prior/updated values.

4.3 Target User Interface Design

- **Unified Workspace:** Replace multi-tab navigation with a clean, workflow-driven single-page dashboard.
- **Input Context:** Group data entry fields logically by operational task rather than mirroring raw database structures.
- **Validation Alerts:** Display inline visual warning indicators for abnormal entries (e.g., task counts exceeding realistic daily limits or excessive late arrivals) prior to final submission.